

BESS Rolling-Horizon Co-Optimization: Mathematical Model & Algorithm

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1 Introduction

This document details the Mixed-Integer Linear Programming (MILP) and Model Predictive Control framework used for Battery Energy Storage System (BESS) co-optimization across Day-Ahead (DA), Ancillary Reserve, and Intraday (ID) markets. Key features:

- Co-Optimization Logic: Efficiently manages participation across Day-Ahead (DA), Intraday (ID), and Ancillary Reserve markets.
- Gate Closures: Accurately models the operational timeline with strict lock-ins for reserves at 08:00 AM and DA auctions at 12:00 PM for the following delivery day.
- Robust Envelopes: Tracks physical battery constraints and safety margins using SoC trajectories and uncertainty bounds.
- Comprehensive Logging & Visualization: Generates a 5-panel figure tracking prices, power dispatch, SoC with envelope shading, and cumulative cash flow.

2 The Market Model

The model maximizes expected net revenue

$$\text{maximize } \sum_{s=0}^{S-1} \text{prob}_s \left[\sum_{t=0}^{H-1} (\text{Rev}(s, t) - \text{DegCost}(s, t) - \text{IDCost}(s, t)) \Delta t + \text{TermGain}(s) \right] \quad (1)$$

incorporating battery degradation costs, ID bid-ask spreads penalties and terminal gains

$$\begin{aligned} \text{Rev}(s, t) &= p_{DA}(t)P_{DA}(t) + p_C(t)C(t) + p_{ID}(s, t)(ID_{dis}(s, t) - ID_{ch}(s, t)) \\ \text{DegCost}(s, t) &= \mu_{deg}(P_{ch}(s, t) + P_{dis}(s, t)) \\ \text{IDCost}(s, t) &= c_{id}(ID_{ch}(s, t) + ID_{dis}(s, t)), \\ \text{TermGain}(s) &= \lambda E(s, H), \end{aligned}$$

where all the variables and parameters will be described below. The complete operational problem is to maximize (1) over the decision variable subject to the constraints below.

2.1 Indices, Sets and Parameters

- Time steps $t \in \{0, 1, \dots, H\}$ (e.g. $H = 480$ steps at 15-minute resolution).
- Scenarios $s \in \{0, 1, \dots, S-1\}$, probabilities prob_s , and number of scenarios S .

- DA prices, capacity prices and ID prices p_{DA} , p_C and p_{ID} .
- Degradation cost μ_{deg} and bid-ask spread c_{id} .
- The maximum energy E_{cap} and the maximum power of the battery P_{max} .
- Initial charge E_{init} .
- Charging and discharging efficiency constants η_{ch} and η_{dis} .
- Valuation constant λ for the remaining energy.

2.2 Decision Variables

- $P_{DA}(t)$: day-ahead trading volume.
- $C(t)$: reserved ancillary service (capacity markets)
- $P_{ch}(s, t), P_{dis}(s, t)$: scenario-dependent charging and discharging powers.
- $u(s, t)$: binary variables to avoid simultaneous charging and discharging.
- $ID_{ch}(s, t), ID_{dis}(s, t)$: scenario-dependent intraday buying and selling decisions.
- $E(s, t)$: scenario-dependent energy of the battery.
- Robust envelope variables introduced later on.

2.3 Constraints

Throughout, time is split into 15 minute periods Δt and decision at time t are made just prior to that time. The battery dynamics are described by

$$E(s, t) \in [0, E_{cap}], \quad (2)$$

$$E(0) = E_{init}, \quad (3)$$

$$E(s, t + 1) = E(s, t) + (P_{ch}(s, t)\eta_{ch} - P_{dis}(s, t)/\eta_{dis}) \Delta t, \quad (4)$$

$$P_{ch}(s, t) \in [0, P_{max}], \quad (5)$$

$$P_{dis}(s, t) \in [0, P_{max}], \quad (6)$$

$$P_{ch}(s, t) \leq u(s, t)P_{max}, \quad (7)$$

$$P_{dis}(s, t) \leq (1 - u(s, t))P_{max}. \quad (8)$$

The last two constraints mean that $P_{dis}(s, t)$ and $P_{ch}(s, t)$ cannot both be strictly positive.

We operate the battery by participating in capacity markets (ancillary services), day-ahead (DA) energy markets, and intraday (ID) markets:

- **Ancillary Reserve Gate Closure:** At 08:00 AM, capacity market positions $C(t)$ are locked for the upcoming delivery day. Capacity markets operate in 4 hour blocks, so $C(t)$ are constant within these blocks. Taking the position C , we have to deliver $\alpha(s, t)C(t)$ amount of power, where the activation ratio $\alpha(s, t) \in [-1, 1]$ is only known after the period (we assume that we must deliver, in practice one pays a high penalty when not doing so).
- **Day-ahead auctions:** At 12:00 PM, DA energy schedules $P_{DA}(t)$ are locked for the upcoming delivery day.
- **Intraday markets:** Every 15 minutes, ID adjustments $P_{ID}(s, t)$ are optimized.

In particular, we have fixed profiles $C(t)$ and $P_{DA}(t)$ for times t for which the corresponding auctions are already closed. The net power flow under random activation $\alpha(s, t) \in [-1, 1]$ of the capacity is given by

$$P_{dis}(s, t) - P_{ch}(s, t) := \alpha(s, t)C(t) + P_{DA}(t) + ID_{dis}(s, t) - ID_{ch}(s, t) \in [-P_{max}, P_{max}]. \quad (9)$$

Robust Envelopes (Recourse and Induced Constraints)

To guarantee that the battery can safely fulfill worst-case activations of ancillary services, we introduce robust energy envelopes $E_{max}(t)$ and $E_{min}(t)$. These are the largest and smallest energies in $[0, E_{cap}]$ for which there exist auxiliary deterministic ID schedules $P_{ub}(t)$ and $P_{lb}(t)$ such that

$$E_{max}(t) + ((C(t) + P_{ub}(t))^+ \eta_{ch} - (C(t) + P_{ub}(t))^- / \eta_{dis}) \Delta t \leq E_{max}(t + 1) \quad (10)$$

$$E_{max}(t) + ((-C(t) + P_{ub}(t))^+ \eta_{ch} - (-C(t) + P_{ub}(t))^- / \eta_{dis}) \Delta t \geq E_{min}(t + 1) \quad (11)$$

$$E_{min}(t) + ((C(t) + P_{lb}(t))^+ \eta_{ch} - (C(t) + P_{lb}(t))^- / \eta_{dis}) \Delta t \leq E_{max}(t + 1) \quad (12)$$

$$E_{min}(t) + ((-C(t) + P_{lb}(t))^+ \eta_{ch} - (-C(t) + P_{lb}(t))^- / \eta_{dis}) \Delta t \geq E_{min}(t + 1) \quad (13)$$

$$P_{DA}(t) + P_{ub}(t) \leq P_{max} - C(t), \quad (14)$$

$$P_{DA}(t) + P_{lb}(t) \leq P_{max} - C(t), \quad (15)$$

$$P_{DA}(t) + P_{ub}(t) \geq -P_{max} + C(t), \quad (16)$$

$$P_{DA}(t) + P_{lb}(t) \geq -P_{max} + C(t), \quad (17)$$

where $x^+ = \max(0, x)$ and $x^- = \max(0, -x)$. We add explicitly the constraints

$$E(s, t) + [(C(t) + P_{DA}(t) + ID_{dis}(s, t) - ID_{ch}(s, t))^+ \eta_{ch} - (C(t) + P_{DA}(t) + ID_{dis}(s, t) - ID_{ch}(s, t))^- / \eta_{dis}] \Delta t \leq E_{max}(t + 1), \quad (18)$$

$$E(s, t) + [(-C(t) + P_{DA}(t) + ID_{dis}(s, t) - ID_{ch}(s, t))^+ \eta_{ch} - (-C(t) + P_{DA}(t) + ID_{dis}(s, t) - ID_{ch}(s, t))^- / \eta_{dis}] \Delta t \geq E_{min}(t + 1), \quad (19)$$

$$E(s, t) \in [E_{min}(t), E_{max}(t)]. \quad (20)$$

Note that the robust envelope variables $E_{max}(t)$, $E_{min}(t)$, $P_{ub}(t)$ and $P_{lb}(t)$ become decision variables in the model.

3 Rolling Horizon and Model Predictive Control

We apply the rolling horizon technique. The problem (1) is solved at time $t = 0$, the decisions locked at initial times are applied, and the state of charge is updated for the next step. Repeating until the end of the simulation window, the problem (1) is solved with a new horizon after updating the data.

1. **Initialization:** Load forecasted DA prices, simulate ID prices and ancillary activations.
2. **Gate Closure Enforcement:** At 08:00 AM (Reserves) and 12:00 PM (DA) on D-1, lock market capacity and DA commitments for the upcoming delivery window.
3. **Optimization Solve:** Execute the MILP solver, extract optimal controls for the current step, and apply internal shifted warm-starts to accelerate convergence.
4. **State Propagation:** Roll the validated physical state forward to the next clock tick.

The problem is solved using model predictive control (MPC) and programmed in Python utilizing Pyomo and HiGHS.